Using Drupal and DITA OT to Create a PDF File

This post shows you how to use Drupal and DITA OpenToolkit to create a PDF file. It uses Views in Drupal and the REST UI to create a json file that will be used to create a bookmap and markdown files. These files will be used by Open Toolkit to create a PDF file.

Prerequisites: Drupal, DITA OT, Python

Make sure that the following modules are installed in Drupal:

drupal_web_services.png

Drupal Modules

For DITA Open Toolkit, you can use markdown files instead of DITA files. For this post we will use markdown files, which are much easier to work with. DITA files require that you are familiar with fairly complex tagging. Markdown is much easier to use. See the following website for a markdown cheatsheet: Markdown Cheat Sheet

Creating Dita Bookmap and Markdown Content Types

We will create a Dita bookmap and two markdown files. First, let's make two new Content Types in Drupal.

For the first content type, perform the following steps:

1. Select **Structure > Content Types > Add Content Type** from the Administrator menu.
2. For the Name, enter: **markdown**.
3. For Description, enter: "Use markdown for creating markdown file."
4. Add the fields as shown below. You can Re-use an existing field for Tags drupal_manage_fields.png
5. Edit the Body field and make sure Body is set to allow Plain text.

For the second content type, perform the following steps:

1. Select **Structure > Content Types > Add Content Type** from the Administrator menu.
2. For the Name, enter **Dita Bookmap**.
3. For Description enter: "Use Dita Bookmap for creating a Dita bookmap."
4. Add the fields as shown below. You can Re-use an existing field for Tags.drupal_manage_fields.png
5. Edit the Body field and make sure Body is set to allow Plain text.

Creating Markdown Files in Drupal

Next, we will create the two markdown files. For the first file, perform the following steps:

1. Select **Content > Add Content > markdown**.
2. For the Name, enter: **Test markdown page 1**
3. Enter the following information:

drupal_markdown1.png

For the second file, perform the following steps:

1. Select **Content > Add Content > markdown**.
2. For the Name, enter: **Test markdown page 2**
3. Enter the following information:

drupal_markdown2.png

Creating Bookmap File in Drupal

Let's create the bookmap file next.

1. Select **Content > Add Content > Dita Bookmap**..
2. For the Name, enter: **Test Bookmap**
3. Enter the following information:

drupal_test_bookmap.png

For Body field, enter the following information:

```
<?xml version='1.0' encoding='utf-8'?><!DOCTYPE bookmap PUBLIC "-//OASIS//DTD DITA BookMap//
<bookmap id="test1">
<booktitle>
<booklibrary>Test Document</booklibrary>
<mainbooktitle>Test Document</mainbooktitle>
<booktitlealt>By: Eric Strausbaugh</booktitlealt></booktitle>
<frontmatter>
<booklists><toc/></booklists>
</frontmatter>
<chapter href="test1.md" format="markdown">
 <topicref href="test_page2.md" format="markdown"></topicref>
</chapter>
</bookmap>
```

Creating a Drupal View

The following step shows how to make a view using the dita_test tags that we applied above.

1. Select **Structure > Views > Add View**.
2. Enter **dita_test** for the View Name.
3. Select **Create Page** and use the default settings.
4. Click on **Provide a Rest Export** and chose **api/dita_test** as the Rest Export Path.

5. Click **Save and Edit**.
6. Click on the **Edit** button for the dita_test view.
7. Click on **Add** under Filter Criteria.
8. Click on **Tags** (field_tags).
9. Click **Apply**.
10. Click **Apply and Continue**.
11. For Operator, select Is one of and enter dita_test.
12. Click **Apply**.
13. Click on the **Rest export display**.
14. Under Formats, click on **Settings**.
15. Select **json** as the Accepted request formats and click **Apply**.
16. Click Save.

The Preview should look similar to the following:

drupal_api.png

Creating Bookmap File and Markdown Files

The following steps will create the bookmap file and markdown files from the http://localhost/api/dita_test path by using a Python script:

1. Create a directory (e.g., python_drupal).
2. Change directory to the directory that you created.
3. Create a file named json_to_dita.py.
4. Paste the following contents into the file. Then, run `python3 json_to_dita.py`

```
import json
import requests
import os

def get_json_from_url(url):
    try:
        response = requests.get(url)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)
        return response.json()
    except requests.exceptions.RequestException as e:
        print(f"Error fetching URL: {e}")
        return None
    except json.JSONDecodeError:
        print("Response is not valid JSON")
        return None

def save_json_to_file(data, filename):
    with open(filename, 'w') as f:
        json.dump(data, f, indent=4)
```

```
# Example usage:
url = "http://localhost/api/dita_test"
data = get_json_from_url(url)


def json_open():
    with open('data.json', 'r') as file:
        data = json.load(file)
        length= len(data)
        #print(data)
        print(length)
        for x in range(length):
            print(data[x]['vid'][0]['value'])
            print(data[x]['body'][0]['value'])


length= len(data)
#print(data)
print(length)
for x in range(length):
    directory_name = str(data[x]['field_tags'][0]['target_id'])
    # Create the directory
    try:
        os.mkdir(directory_name)
        print(f"Directory '{directory_name}' created successfully.")
    except FileExistsError:
        print(f"Directory '{directory_name}' already exists.")
    except PermissionError:
        print(f"Permission denied: Unable to create '{directory_name}'.")
    except Exception as e:
        print(f"An error occurred: {e}")
    file1 = open(directory_name + "/" + data[x]['field_name'][0]['value'], "w")
    file1.write(data[x]['body'][0]['value'])
    print(data[x]['vid'][0]['value'])
    print(data[x]['body'][0]['value'])
    print(data[x]['field_name'][0]['value'])
    file1.close()  # to change file access modes
```

This creates the bookmap and markdown files.

Creating PDF File or HTML5 Files

Within the directory with the bookmap and markdown files, run:

```
~/dita-ot-x.x/bin/dita --input=test_bookmap.ditamap --format=pdf --output=out
```

or

```
~/dita-ot-x.x/bin/dita --input=test_bookmap.ditamap --format=html5 --output=out
```

NOTE: DITA-OT must be installed to run the commands above. See https://www.dita-ot.org/dev/topics/installing for installation instructions.